My grades for **Final exam**

Q1

5 / 5

What are RDF (semantic) triples? Using your own example, illustrate (ie. with diagrams) how they can be used to construct a knowledge graph ('KG').

RDF triples are known as Resource Description framework triples. They are a standard way to represent information in a structured format especially when it comes to the Semantic Web.

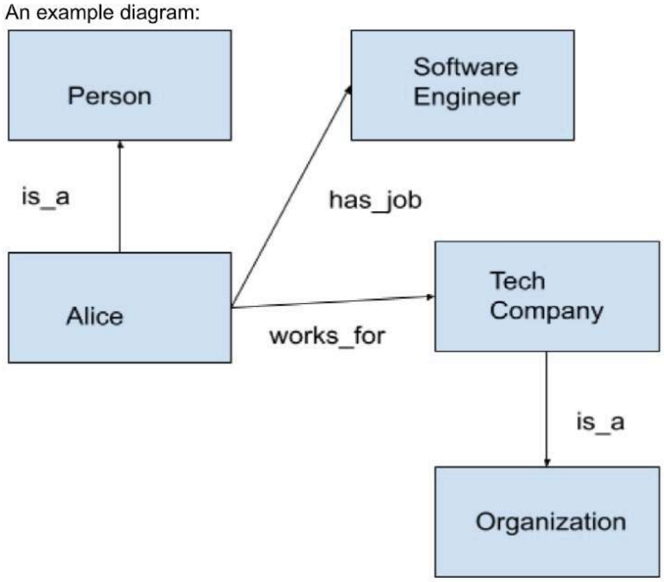
Three components:

- Subject
- Predicate
- Object

Examples:

- (Alice, is_a, Person)
- (Alice, has_job, Software_Engineer)
- (Alice, works_for, Tech_Company)
- (Tech_Company, is_a, Organization)

An example diagram:



Each node in a KG/Knowledge Graph represents an entity and the edges, the relationships between. The relationships are the predicates as described above. This allows us to represent complex entity relationships in a structured way making the data easier to understand and query.

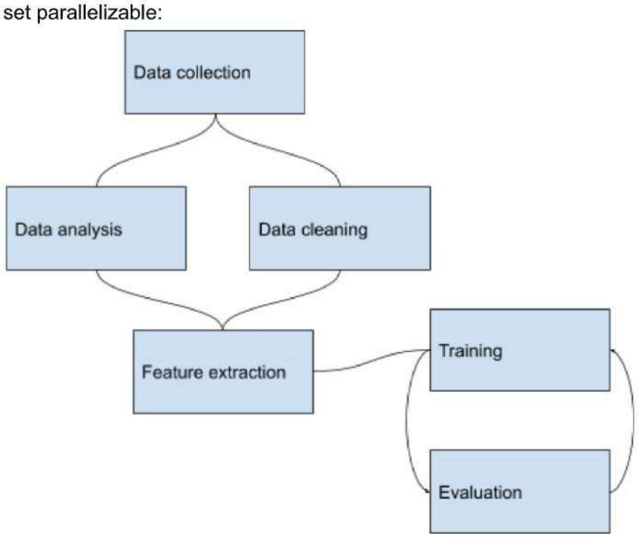
Q2

3 / 5

In many data processing pipelines, several steps can be carried out at the same time (speeding up processing), while other steps cannot. How would we describe such a pipeline at a higher level (of specification), and how, at a lower level? Illustrate using a simple example, and provide short explanations.

Q2

In a data processing pipeline, there are 2 types of steps: Sequential and parallelizable tasks. Sequential means that the task depends on the previous task output to completely finish so the current can start. Parallelizable means that there are one or more tasks that do not depend on the current task to finish to run. At a high level we can describe the pipeline using a DAG and set parallelizable:



-2 missing example

At a high level we can use a gantt chart to showcase the overlap in steps

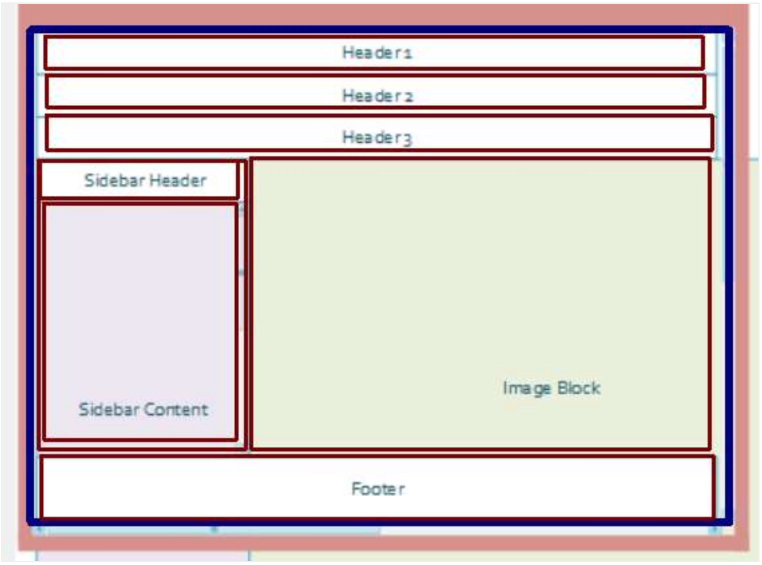
t	1	2	3	4
Data Collection				
Data analysis				
Data cleaning				
Feature extraction				
Training				
Evaluation				

A gantt chart can have subsequent steps and detail which steps can be concurrent with other steps in the pipeline

Q3

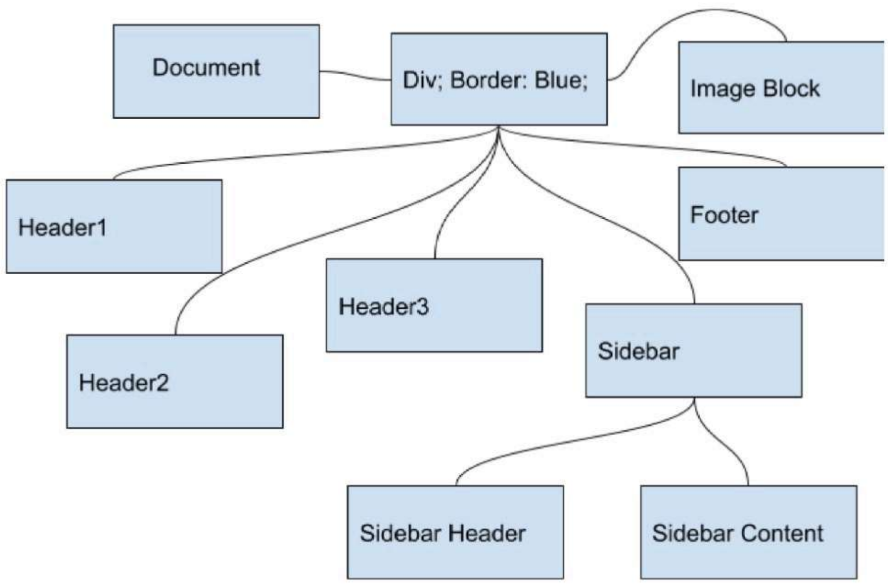
5 / 5

Consider the following UI layout, where a blue container (eg a 'div' in HTML) holds the other elements (in dark red), in a 'nested' layout that is typically how UI gets laid out - it can be regarded as a form of data specification.



1. Draw a simple tree that shows the layout hierarchy. For anything not named (hint - there is one such!), you can make up a name for it.
2. Describe the layout in syntactically valid JSON, using "layout" as the main (top-level) key.

Draw a simple tree that shows the layout hierarchy. For anything not named (hint - there is one such!), you can make up a name for it.



Describe the layout in syntactically valid JSON, using "layout" as the main (top-level) key.

```
{
    "document": {
        "Header1": "Header1",
        "Header2": "Header2",
        "Header3": "Header3",
        "Sidebar": {
            "SidebarHeader": "SidebarHeader",
            "SidebarContent": "SidebarContent"
        },
        "ImageBlock": "ImageBlock",
        "Footer": "Footer"
    }
}
```

Q4

5 / 5

1. In one word - what do we collect, clean, store data for?
2. Using your HW2 through HW5 as examples, explain in a couple of lines for each, how you made '1.' above, happen.

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1. Insights
2. In HW2 we simulated the database structure of a simplified youtube-like web application. One of the queries was to fine the unique users whose channel has more than a 100 paid subscribers and the channel is created on 1/1/23. We were able to retrieve metrics and view channel insights using sql queries over structured data. Similarly in HW3 we ran queries to store spatial data and ran queries to retrieve and calculate data that can be abstracted from it. Examples include creating a bounding box, and finding the top k nearest points from a home point. In HW4 we were able to visualize the spatial data using MongoDB Atlas and show points within a certain boundary which can be a polygon of any shape whether it be a triangle, square or an n-gon polygon. Finally in HW5 we created a machine learning model to gather the subtle insights from the data we provide it to then intelligently classify any new data that we provide it. by doing so we demonstrate that gathering insights does not necessarily need to be done manually. By plotting the weights on a heatmap for the last layer of the machine learning model we are able to visualize an approximate shape of what is classified as 1 and what is classified as a 0.

Q5

5 / 5

In ML, why are loss functions important (what do they help with)?

What is the most important part of a neuron in an artificial neural network? WHY?

The UI in Q3 above, can be coded up using CSS (eg using divs and the 'flexbox' layout). But we could make genAI (eg ChatGPT) output such CSS for us, rather than writing it ourselves. What would be the training data, for this to become possible?

Loss functions are crucial due to them being the quantification of the performance of the model. Using the metrics from the loss function we are able to mathematically fine tune the model in an automated fashion to be more accurate in predicting the answers to the data we provide. The loss function is usually used during the training process to adjust the weights and is done so with an optimization process like gradient descent. The goal is to minimize the loss from the loss function. By doing so we reach a higher accuracy score.

The most important part of a neuron in an artificial neural network is the activation function. The reasons are as follows: activation functions introduce non-linearity: regardless of how deep the neural network is, it would behave like a shallow neural network if the results are linear functions. Example: single layer perceptron The activation function controls the output and decides if the neuron should be activated or not by calculating the sum and bias. activation functions can also normalize the output into a range like 1 and 0 (ReLU) or -1 and 1 (Sigmoid) Different functions have different advantages and the correct choice can greatly affect the overall performance of the network.

For generative AI such as ChatGPT to output CSS the training data can in be one of 2 forms: Input: Text; Output: Text Input: Image; Output: Text

Incorrect MC
Containers-C

Going the route of Text-to-Text is most common as the transformers used in the underlying model are industry standard. The content of the input could be descriptions of what the CSS should look like and the output as CSS code pulled from public repositories or websites.

Correct 1

Going the image route is a bit trickier as an image can either be analyzed using an Image-to-Text model like OpenAI's CLIP then fed into a Text-to-Text model or the image can be directly encoded into an embedding vector and used as an input. The data would be similar to the Text-to-Text route but would instead be Images as input and CSS as the output.

Q6

2 / 2

For a number of years, databases were of the relational (tables) and NoSQL (k:v, documents, columns, graph) type. Now we have a new category - 'vector DBs', that are for carrying out 'similarity search' (where we query for data, without using SQL etc., but more 'loosely'). Explain in two or three sentences, how these queries/DBs work.

Vector databases store data in a vectorized (list of numbers) format, often in high-dimensional spaces. These databases are designed for similarity search, where you query by providing a vector, and the database returns the most similar vectors in the database based on some distance metric (like cosine similarity or Euclidean distance). This makes them useful for cases like recomendation systems, image or text recognition, and other machine learning tasks like text embeddings where data is represented as vectors in high-dimensional spaces.

Q7

2 / 2

To query data, we mostly still use syntax-based languages such as SQL, Python, etc. Name, and discuss, TWO up-and-coming alternatives to this (hint: one of these is already in use, rather widely!).

Natural language based querying where a vector database with text embeddings can eb queryed with embeddings from the query. By using vector similarity search users can retrieve relevant data and push it as in-context data to an LLM for context based queries.

Graph based queries: by modeling the relation between entities a graph can be created which can allow users to query data based on said relationships more intuitively. GraphDB's like Neo4j and Neptune are used for relational based reccomendation systems and anomaly/fraud detection systems by modeling relations between data such as movies or fraud incidents (respectively)

Q8

2 / 2

"From BI to AI" - what is the connection (similarity) between BI (data warehousing), and AI (a colloquial substitute for ML/DS)? Be specific, and answer in 2 or 3 lines.

BI is where data analysists do their magic in finding relavant metrics within the data and by doing so can relay that information to the ML Engineers/Data Scientists to build models that utilize that data in the most optimal way possible. by analyzing and filtering out irrelevant data, predictions from ML models can be much more accurate.

Q9

1 / 2

Large-scale data storage and querying have evolved, from being entirely centralized, to being entirely distributed. 'MCC' is how we currently "do distributed". In this context (MCC for data access), what does 'serverless' mean?

In the context of large-scale data storage and querying, 'MCC' refers to Multi-Cloud Computing, which involves distributing data and processing across multiple cloud service providers to improve scalability, availability, and fault tolerance. Serverless, in this context, refers to a computing model where the cloud service provider manages the underlying infrastructure, and users only need to focus on writing and deploying their code or queries.

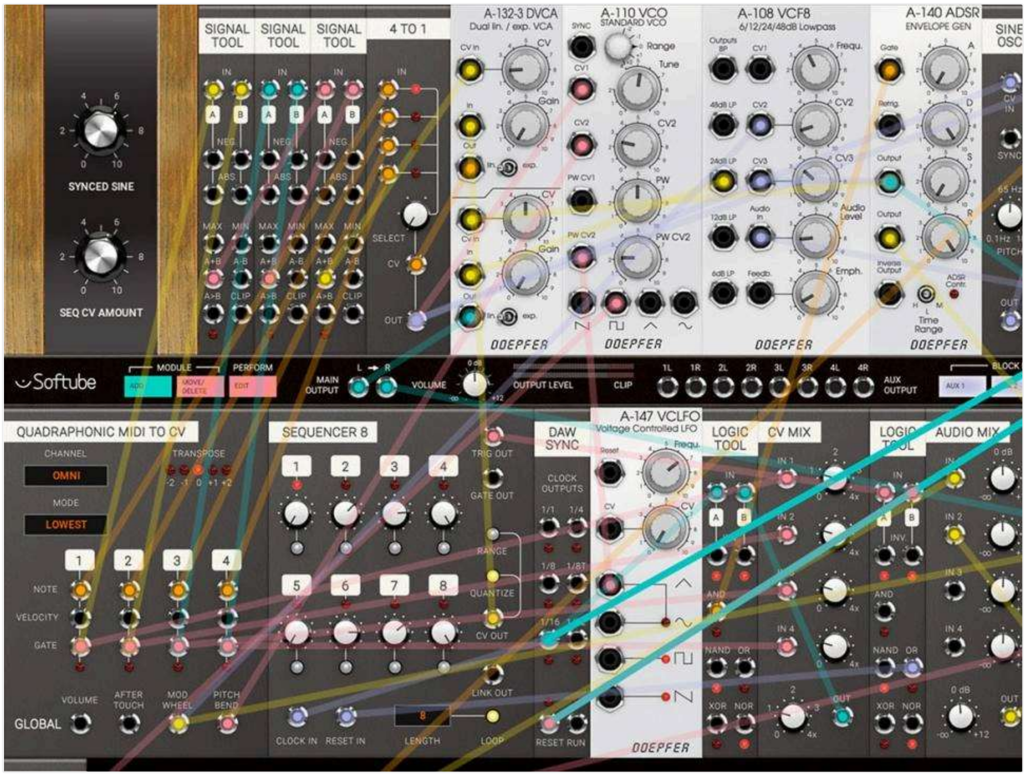
Serverless computing for data access means that users do not have to manage servers, networking, or storage infrastructure for their data processing tasks. Instead, they can leverage cloud-based services like AWS Lambda, Google Cloud Functions, or Azure Functions to execute their queries or data processing tasks. These services automatically scale based on the workload, and users only pay for the compute resources consumed during the execution of their tasks.

In serverless computing users write event based logic.

Q10

0 / 2

One way to synthesize complex audio is to hook up nodes (eg. https://noisecraft.app/623) or modules (like in the graphic below). How does this relate to data handling? In other words, in a few sentences - how do we conceptualize data pipelines? Hint: think of three 'stages' we discussed in class.



This question wasn't answered

Q11

2 / 2

LLMs such as GPT-4 are 'pretrained' (that is the 'P') with a large amount of language data. For more precise answers, they need to be 'fine tuned' with specific domain-related (eg medical, legal...) language, OR, be chained to a custom DB that can be accessed by the GPT. In this context, what is the 'OPL stack'? Describe in your own words, sticking to what we discussed in class (high level description is fine).

OPL is OpenAI, Pinecone, and langchain. By using OpenAI we can access GPT3.5 or GPT4 and also the embedding models to embed our text. Pinecone is a vector database that can store our embeddings for similarity searching and contextual prompt injection for in-context learning. Langchain is the python/javascript framework that ties everything together by providing frameworks to create pipelines to ingest and retrieve data which is used to query a LLM like openai's GPT4.

Q12

0 / 2

AI techniques can be classified as being one of these four types: heuristic/brute-force, reinforcement learning (RL), rule-based, or connectionist ("ML"). Of these four, which is data-driven? Explain how data is used to impart intelligence to the machine? Briefly outline the steps involved.

This question wasn't answered

Q13

0 / 2

Palantir is an ML-oriented company that works with law enforcement, etc. Recently they announced this:

vice.com/en/article/qjvb4x/palantir-demos-ai-to-fight-wars-but-says-it-will-be-totally-ethical-dont-w...

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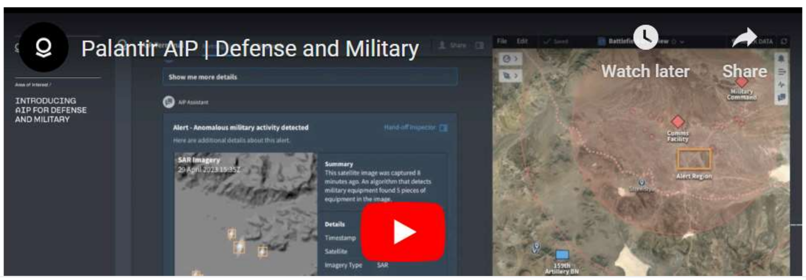
IMAGE VIA YOUTUBE.

Palantir, the company of billionaire Peter Thiel, is launching Palantir Artificial Intelligence Platform (AIP), software meant to run large language models like GPT-4 and alternatives on private networks. In one of its pitch videos, Palantir demos how a military might use AIP to fight a war. In the video, the operator uses a ChatGPT-style chatbot to order drone reconnaissance, generate several plans of attack, and organize the jamming of enemy communications.



In terms of 'GPSETC', **what would be considered alarming** about this?

Also, here is another recent news item (related to this: data can be used to train a system to generate alternative data, ie. to create 'generative AI'):

TikTok Is Developing AI-Generated Video Disclosures as Deepfakes Rise

By Kaya Yurieff

Some viral TikTok videos may soon show a new type of label: that it's made by AI.

The ByteDance-owned app is developing a tool for content creators to disclose they used generative artificial intelligence in making their videos, according to a person with direct knowledge of the efforts. The move comes as people increasingly turn to AI-generated videos for creative expression, which has sparked copyright battles as well as concerns about misinformation.

What 'E'thical and 'S'ecurity issues do deepfakes pose? Be original in your answer!

This question wasn't answered